Machine Learning

Course Overview
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Learning Objectives

- Introduction to machine learning
- Mathematical foundations of ML
- Practical development of ML

Course canvas page

- Syllabus, schedule, policies

- Join intro discussion

Course format

- Class
  - Attendance & participation are expected (**5%**)

- Assignments (**45%**)
  - Start early, submit on time

- Midterm exam (**15%**)

- Final project (**35%**)
  - Semester long effort, pick a topic, form your team!

Reach out!

- Join office hours
- Post on canvas/**discussion**
- Reach us on course email: ml.spring26@gmail.com
- Suggestions welcome

Project Roadmap

- Join a team and brainstorm project ideas
- Draft a formal project proposal
- Develop a proof of concept and code demo
- Start drafting your project paper
- Refine your project and work on the project deliverables
- Present your project in class
- Finalize your project deliverables
- Submit your final project

Grading policies

- Group project **graded individually** based on contribution
  - Being an effective team player counts towards your grade
  - Indicate individual contribution (percentage & description) on every submission
  - Split work evenly

- Late Policy
  - First late day penalty - 10%
  - Second late day - 25%
  - Third late day - a grade of zero will be given

Course policies

- **Accessibility**
  - For special accommodation, please contact me during the first week of classes

- **Academic Integrity**
  - Abide by **Graduate Honor Code**
  - Zero-tolerance for plagiarism/other forms of cheating
  - Academic dishonesty reported to Graduate Honor System
  - Collaboration encouraged but must be documented
  - Solutions must be your own work & must be written independently
  - Please cite external sources of code, ideas, or language

- We will adhere to **Virginia Tech's Principles of Community**

- Grading scheme
  [0-60] F, [60-62] D-, [63-66] D, [67-69] D+, [70-72] C-, [73-76] C, [77-79] C+, [80-82] B-, [83-86] B, [87-89] B+, [90-92] A-, [93-100] A

Textbooks

- We are not using a particular textbook

- List of useful books on canvas/pages

Prerequisites

- [ECE 3514 or CS 2114]: grade C or higher
- [STAT 3704, STAT 4105, STAT 4604, STAT 4705, STAT 4714, or CMDA 2006]: grade C or higher
- Significant programming experience:
  - written programs of 100s of lines of code
- Python, data structures, algorithms, statistics
  - Review material on canvas/syllabus

GTAs

- Jiazhen Hu
- Siyi He